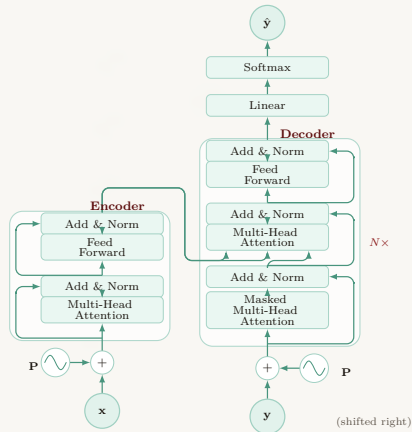


Capable models, costly deployment

The Transformer is the dominant architecture for language and sequence modelling. The same scale that gives these models their capability also makes them expensive to deploy, in memory, latency, and energy, and often impractical on resource-limited hardware.

Model compression narrows the gap between training-scale capability and deployment-scale constraints. To compress a model in a principled way, we first need to know what it is built from, and in what units its cost is measured.



The remainder of Part I builds exactly this vocabulary: the network, the Transformer, the cost metrics, and the search method.